

مجال مغناطيسي

A region around a moving charge

A region around a magnet

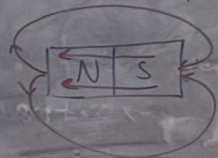
A region around a current carrying conductor
wire having current

B lines

Magnet: Any magnet consists of N and S magnetic poles

Strong: at ends - at poles

Weak: at the center

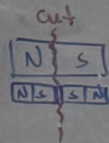


B lines are directed from N into S (out)

B lines " " " S into N (in)

Magnetic field $\rightarrow B$

if we cut a magnet into 2 parts



result: we get two smaller magnets

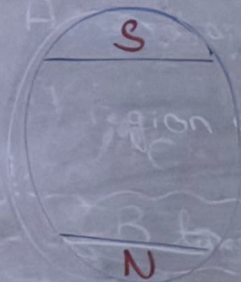
* There is no isolated magnetic poles
" " " single " "

* There is a pair of magnet poles

مجال مغناطيسي

Magnetic field $\rightarrow B$

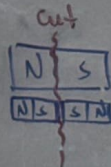
Magnetic of Earth



North geographic pole

South geographic pole

if we cut a magnet into 2 parts



result: we get two smaller magnets

* There is no isolated magnetic poles
" " " single " "

* There is a pair of magnet poles

N pole of compass is directed to N geographic of earth
" " " " " " " S magnetic of earth

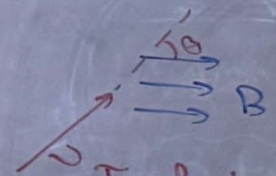
29/12/2019

Magnetic Force $\rightarrow F_B$

$$F_B = q \vec{v} \times \vec{B} \rightarrow F_B = q v B \sin \theta$$

between v and B

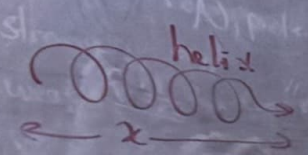
F_B is directed Perpendicular to v and B .



Type of motion ①
Straight

$\theta = 0^\circ$ or $\theta = 180^\circ$:
Same Opposite

$$F_B = 0$$



Helix

② $0^\circ < \theta < 90^\circ$: $F_B = q v B \sin \theta$



Circular

③ $\theta = 90^\circ$: $F_B = q v B$ (maximum)

بسم الله الرحمن الرحيم

+x right →

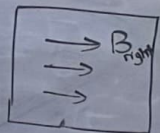
-x left ←

+y up ↑

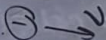
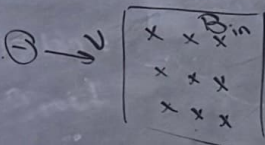
-y down ↓

+z out •

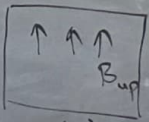
-z in x



F_B at -z or into a paper

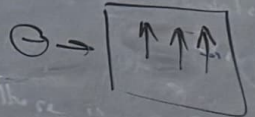


F_B at -y down



$F_B = 0$

result



$F_B = -z$ (into)

F_B in -z $q = +$
 v in +x

B at -y